

Essential Summary

Generalized Padeé Approximation Method for Homoclinic Orbits of Strongly Nonlinear Autonomous Oscillators

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1 Background and Motivation

The determination of homoclinic orbits is central to understanding global bifurcations, chaos, and solitary wave solutions in nonlinear dynamics. Existing Padeé-based methods share a limitation: approximants are restricted to polynomial numerators and denominators, reducing flexibility for exponential-decay homoclinic behavior.

This paper introduces the **generalized Padeé approximation (GPA) method**, extending numerator and denominator from polynomials to series of *any* function type. This unifies prior Padeé-type variants as special cases. A novel approximant using hyperbolic functions is constructed for solving homoclinic orbits of strongly nonlinear autonomous oscillators.

2 Core Innovation

2.1 The Generalized Padeé Definition

Let $\hat{H}_m = \{P : P(z) = \sum_{i=0}^m a_i g_i(z)^i\}$ and $G(m, n) = \{R = P/Q : P \in \hat{H}_m, Q \in \hat{H}_n \setminus \{0\}\}$. The GPA $[L/M]_f$ is $P_L/Q_M \in G(L, M)$ satisfying:

$$f(z) - \frac{P_L(z)}{Q_M(z)} = \mathcal{O}(z^{L+M+1}). \quad (1)$$

When $g_i(z) = z^i$, the GPA reduces to the classical Padeé approximant.

2.2 Homoclinic Orbit via Hyperbolic Functions

For $\ddot{x} + g(x) = 0$ with saddle $H(h, 0)$, the power series $x(t) = \sum_{i=0}^{\infty} a_i t^i$ is obtained by substitution. The initial amplitude a is from $\int_h^a g(x) dx = 0$. The key step: construct a GPA using $\text{sech}^i t$ that automatically satisfies the boundary condition $x(\pm\infty) = h$:

$$\text{GPA}[L/M] = \frac{\sum_{i=0}^L \alpha_i \text{sech}^i t}{1 + \sum_{i=1}^M \beta_i \text{sech}^i t}. \quad (2)$$

The $L + M + 1$ coefficients $\{\alpha_i, \beta_i\}$ are determined by Taylor-expanding at $t = 0$ and matching with the power series $\{a_i\}$ — a purely algebraic procedure.

3 Key Results

3.1 Example 1: Strongly Asymmetric Helmholtz–Duffing

$$\ddot{x} + c_1 x + c_2 x^2 = 0, \quad c_1 = -3, \quad c_2 = 100. \quad (3)$$

With $L = M = 2$, GPA[2/2] gives:

$$\text{GPA}[2/2] = \frac{0.005 \operatorname{sech} t + 0.07 \operatorname{sech}^2 t}{1 + 0.611111 \operatorname{sech} t + 0.055555 \operatorname{sech}^2 t}. \quad (4)$$

The phase portrait matches Runge–Kutta with excellent agreement despite $c_2 = 100 \gg 1$.

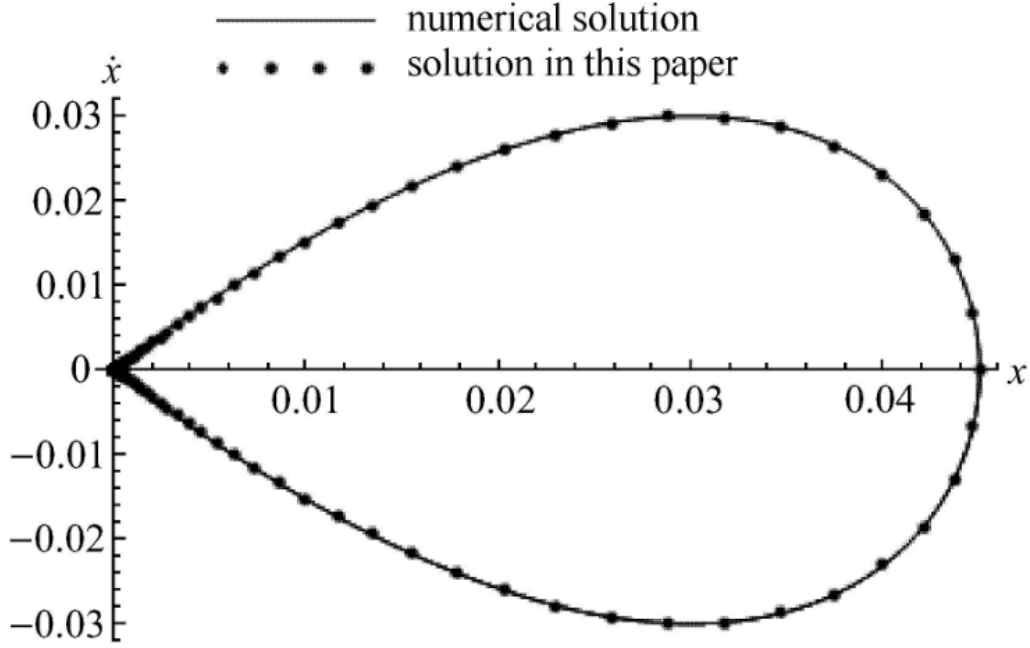


Figure 1: Homoclinic orbit for $c_1 = -3, c_2 = 100$ (Original Fig. 1). Solid: Runge–Kutta. Dotted: GPA[2/2].

3.2 Example 2: Cubic–Quadratic Helmholtz–Duffing

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 = 0, \quad c_1 = -2, \quad c_2 = 5, \quad c_3 = 15. \quad (5)$$

Both left and right homoclinic orbits are obtained simultaneously with $L = M = 2$, showing the method handles asymmetric nonlinearities without modification.

3.3 Example 3: Duffing–Harmonic Oscillator

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = 0, \quad \lambda = -1, \quad \mu = 1, \quad \nu = 1. \quad (6)$$

With $L = M = 3$, the GPA captures homoclinic orbits of this rational-restoring-force oscillator — a system whose exact solution is not expressible in elementary or classical elliptic functions.

4 Significance

1. **Theoretical unification:** The GPA definition unifies classical Padeé and all quasi-Padeé variants under a single framework.

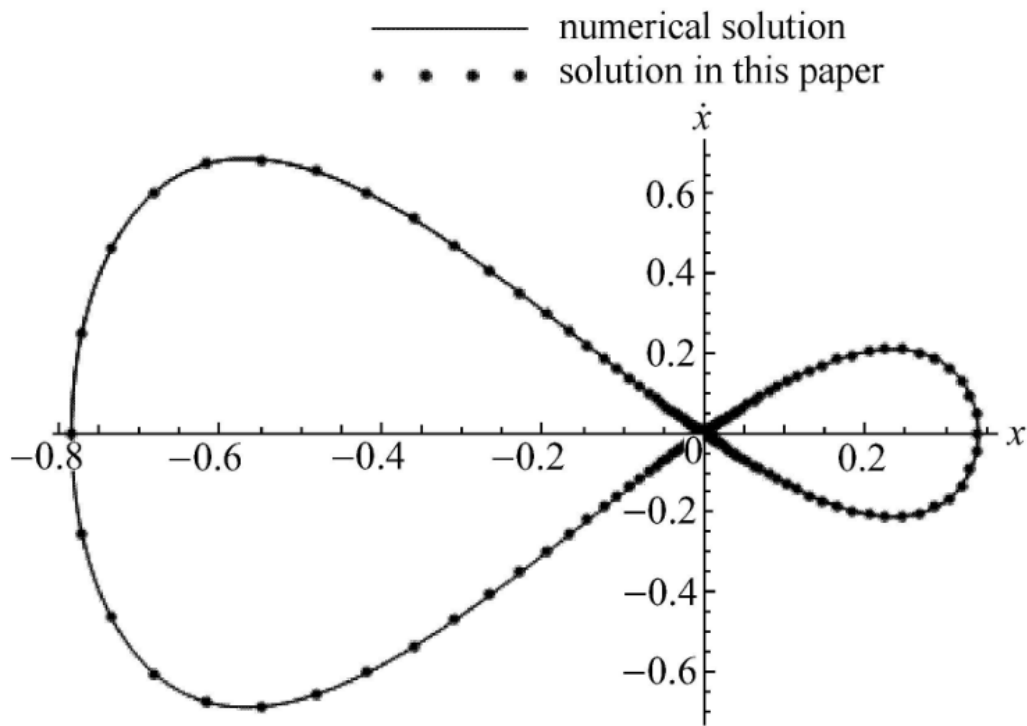


Figure 2: Homoclinic orbits for $c_1 = -2, c_2 = 5, c_3 = 15$ (Original Fig. 2). Solid: Runge–Kutta. Dotted: GPA[2/2].

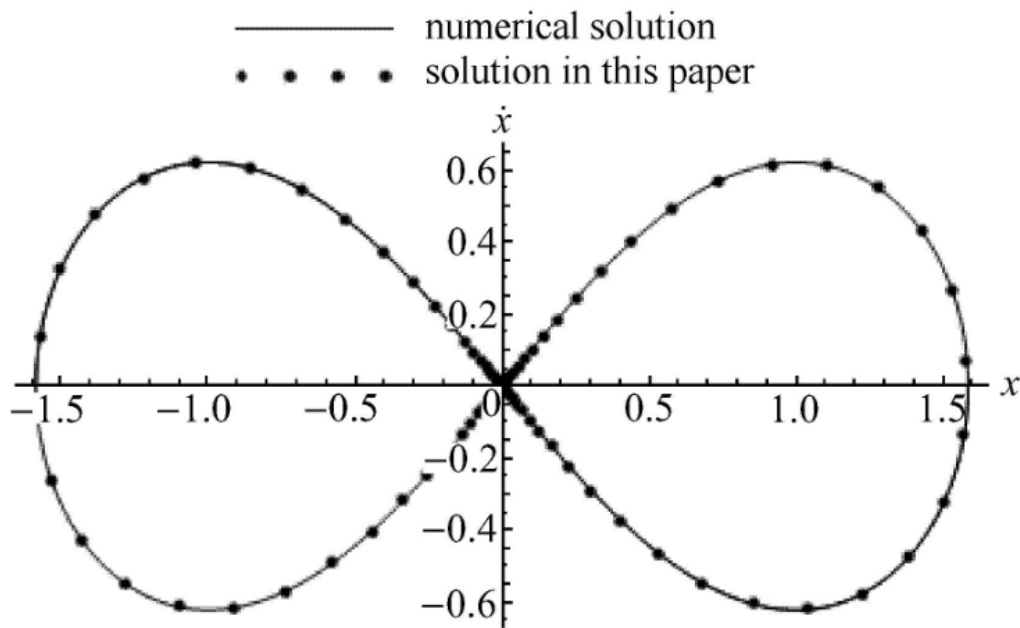


Figure 3: Homoclinic orbits for $\lambda = -1, \mu = 1, \nu = 1$ (Original Fig. 3). Solid: Runge–Kutta. Dotted: GPA[3/3].

2. **Computational efficiency:** $\operatorname{sech}^i t$ has simpler Taylor coefficients than exponentials, reducing matching cost at the same order.
3. **Broad applicability:** Handles polynomial $g(x)$ of arbitrary degree and rational $g(x)$ without modification.

5 Limitations

- Conservative systems only ($\varepsilon = 0$).
- Approximation order chosen empirically; no a priori error bound.
- Single-degree-of-freedom only.

6 Connection to the GHFP/MGHFP Series

This paper and its English version (Chin. Phys. B 2014) and the GHFP/MGHFP series approach nonlinear oscillators via complementary routes: GPA directly constructs homoclinic solutions through hyperbolic-function rational approximants; GHFP/MGHFP uses angular-domain perturbation theory for amplitude–parameter relationships. Both share the mathematical insight of rational function approximation.